

SLOWLAB: Benchmarking Scientific Agents When Experiments Are Slow, Noisy, Costly, and Irreversible

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Abstract

Benchmarks for automated experimental design make experiments instantaneous, free, and infinitely repeatable. Real experiments are none of these. We introduce SLOWLAB, an executable benchmark under four conditions that hold at once: feedback is slow, noise is irreducible, trials are priced in the objective’s own units, and commitments cannot be rolled back. An agent runs a greenhouse — it designs experiments, commits them, waits out a growing cycle, and repeats. Ground truth is a published mechanistic crop model, so the difficulty is not ours to tune, and everything is denominated in net profit per square metre per day. Beside realised regret we report the regret a perfect reasoner would still carry given the same design, computed from an instance distribution no agent sees. The gap separates a campaign that could not have answered the question from one that could and did not, and scores an agent against what was achievable rather than against a baseline we happened to implement.

Across five models, four tasks and 800 episodes, no arm reaches the regret a perfect reasoner would carry having run no experiment at all, and the campaigns fail at the design and at the decision separately. 99% of six-factor screens cannot separate the effects they were commissioned to separate, because the median design replicates two distinct treatments across the whole facility whatever the task needs: under a hard parallelism cap precision and coverage compete for the same physical units, and these agents buy precision. This is the opposite of the failure we expected. On the two higher-dimensional tasks the recommendations then use a twentieth of what even those designs did support, and in 44% of episodes the first cycle of data leaves the recommendation worse than the one the model gave for free. Supplying standard tools moves the two measures in opposite directions — the designs improve and the campaigns end worse in sixteen of twenty cells, because the inference tool is fitted to the agent’s own design, inherits its deficiency, and is deferred to in proportion to how wrong it is. Under these conditions an agent cannot afford the cycle it would take to find that out.

1 Introduction

Language-model agents are increasingly proposed as operators of scientific experimentation: selecting what to run, interpreting what comes back, and deciding what to run next [Boiko et al., 2023, M. Bran et al., 2024, Szymanski et al., 2023]. The claim attached to this work is that such agents will accelerate discovery in domains where experiments are the bottleneck. Whether that claim holds is an empirical question, and answering it requires evaluating agents under the conditions that make experimentation a bottleneck in the first place.

Existing evaluations do not. In the benchmarks available for automated experimental design, an experiment is a function call: it returns immediately, costs nothing, and may be repeated without limit [Gandhi et al., 2025]. Under those conditions the strategy that wins is the one that queries most, and a poor decision is corrected by making another. Neither property survives contact with a laboratory. A commercial tomato crop occupies its greenhouse for seven months and more; a synthesis run consumes reagents and instrument time; a cohort study, once enrolled, cannot be re-enrolled. In these settings the binding constraint is not the agent’s throughput of ideas but the fact that each idea must be paid for in time and material, and that a

*Project: <https://github.com/dingningpei/slowlab-benchmark>

design once committed cannot be revised. An agent optimised against free, instantaneous, repeatable experiments is being selected on a proxy from which the operative difficulty has been removed, and we currently have no executable environment in which to find out whether that selection transfers.

The gap is recognised in the domain literature, which asks for exactly the properties an idealised environment cannot expose—cost-aware performance, constraint-violation behaviour, reproducibility (§2)—and lacks an environment in which to measure them.

SLOWLAB. We introduce an executable benchmark for experimental design under four conditions that hold at once. Feedback is **slow**: a cycle is 210 days, and the number of trials that run concurrently is capped — unevenly, because some factors are set for a whole block of units and others per unit, so replication and coverage compete for the same physical resource. Observations are **noisy** enough that one unit does not settle a question. Experimentation is **costly** in the objective’s own units, since testing a management regime means growing a crop under it and selling whatever results. And a submitted design is **irreversible**: no branch, no checkpoint, no revert. Each condition alone is familiar. Together they withdraw the recourse every existing benchmark assumes, because an agent can neither buy precision and coverage from the same units nor find out which it needed before the season ends.

The conditions define a class rather than a domain — microbial evolution, bioprocess scale-up, materials ageing and dose-finding trials satisfy them too, and Appendix A states what a domain must supply to be admitted. We instantiate the first in controlled-environment horticulture, where a published mechanistic crop model [Jones et al., 1999] coupled to an explicit economic one puts the difficulty outside our control and every quantity in one currency, net profit per square metre per day (§4.2). The agent’s action is a *design*: which factor settings to test, on how many units, and which physical units those are. Lamps and heating serve a whole chamber while planting density and the pruning target are set per hydroponic loop, so an allocation that varies temperature between two loops of one chamber describes an experiment that cannot be performed. Having committed, the agent sees nothing until the cycle ends; after two or three cycles it must recommend. Figure 1 is the whole loop on one page.

Overall, our work makes the following contributions:

1. We introduce SLOWLAB, a *framework for experimental design under slow, noisy, costly and irreversible trials*, in which one harness — protocol, action space, feasibility checker, agent interface — serves a class of domains rather than a single one. A domain enters by supplying a published response model, a cost model denominated in the objective, and a block structure that makes allocation a decision (Appendix A); horticulture is the first, so the optimum is a crop model’s and not ours to tune.
2. We introduce a *decision-theoretic measure of design quality*: the regret a perfect reasoner would still carry given the design, computed from an instance distribution no agent sees. It separates a campaign that could not have answered the question from one that could and did not, contains no baseline of ours, and is the expected value of sample information — a batch Knowledge Gradient (Appendix F).
3. We conduct *the first evaluation of language-model agents under these four conditions jointly*: five models, four tasks, twenty instances each, run with and without standard tools. Over 800 episodes no arm reaches the regret a perfect reasoner would carry having run nothing at all. We expected over-exploration and found the reverse: the median design replicates two distinct treatments across the facility whatever the task needs, leaving 99% of six-factor screens unable to separate the effects they were commissioned to separate, and the recommendations then use a twentieth of what even those designs did support. Supplying a screening tool repairs the designs and makes the campaigns end *worse* in sixteen of twenty cells: one intervention, two measures, opposite directions.

2 Related work

Experimental design benchmarks. Gandhi et al. [2025] is the closest prior work. It scores design quality by expected information gain about the parameter vector—the Bayesian optimal-experimental-design objective, estimated by nested Monte Carlo. We take the same stance, that design quality should be measured from theory rather than from a hand-built checklist. We do not take the same quantity. Ours is a decision-theoretic one: the Bayes risk a design leaves behind on the *recommendation*, not the entropy it removes

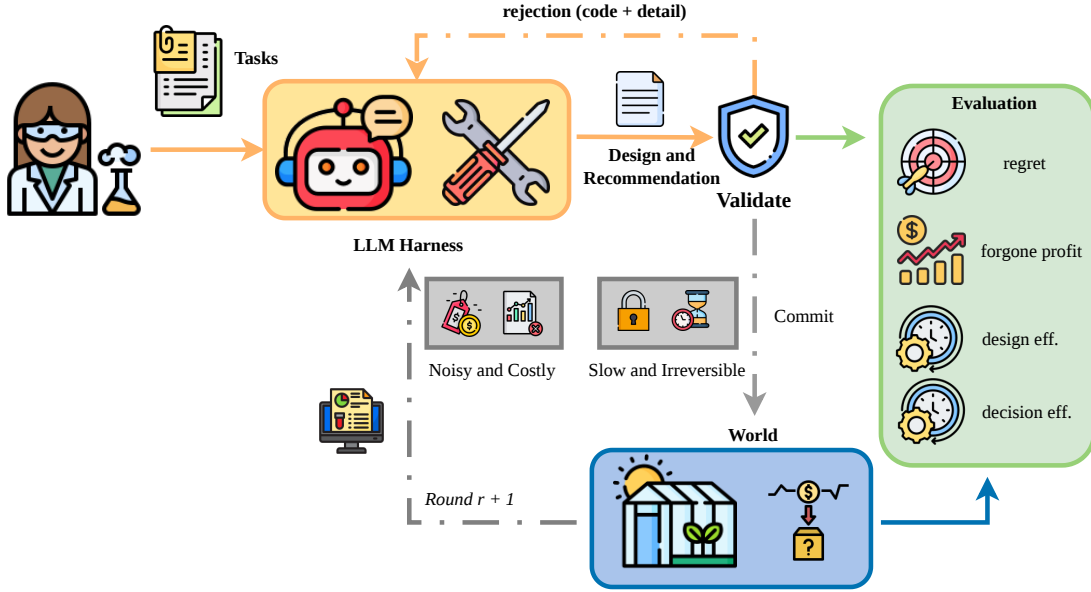


Figure 1: **Overview.** A seed draws a *site*—a parameter vector for the crop model—from a published between-site spread, unknown to the agent. Each round the harness serialises the state, the agent returns a complete design and an interim recommendation, and the design is checked for mechanical feasibility only: a rejection carries a code and a detail and costs no budget, so the agent may revise without penalty. Committing is the irreversible step. Committed units are occupied for a full growing cycle—210 days, so a three-round campaign runs for 630—and the harness returns their values when that cycle ends. What the agent is denied is not sight of the crop but any action to take on it while the units are busy. Units are not interchangeable: some factors are set per chamber and others per loop, so replication and coverage draw on the same physical resource. At the end the campaign is read three ways: the regret its recommendation realised, the profit its experiments forwent, and how much of the room its designs closed and its decisions then used (§3.3).

from the parameters (§3.3.2). A design can pin θ down tightly and spend its budget far from the optimum, which scores well on theirs and badly on ours; their task is model discovery and ours is a decision. And the regime differs: experiments there are instantaneous, free and unboundedly repeatable, with no parallelism constraint and no irreversibility, so the arithmetic behind our main result—a replicate and a new treatment competing for the same physical unit—cannot arise.

Optimal experimental design. The quantity in our second contribution is old. Bayesian experimental design begins with Lindley [1956] and is surveyed by Chaloner and Verdinelli [1995]; its decision-theoretic branch, which scores a design by the Bayes risk it leaves on a *decision* rather than by the entropy it removes from parameters, is Raiffa and Schlaifer [1961] and Howard [1966]. The same object appears in Bayesian optimisation as the Knowledge Gradient [Frazier et al., 2008], beside entropy-search criteria that target the optimum’s location instead [Hennig and Schuler, 2012, Hernández-Lobato et al., 2014]. Estimating any of them is the standing difficulty [Rainforth et al., 2018], and Appendix F reports how ours failed before it worked. What we contribute is not the measure but its use as a benchmark score under conditions — capped parallelism, priced trials, no retries — in which an agent cannot substitute more queries for a better design.

Task validity. Anonymous [2026] trains a model to author executable environments and solve them in self-play, scoring *task* quality by hint-based regret, $r_D(e) = \bar{r}_A(e | h) - \bar{r}_A(e)$: the gain a solver gets from privileged information. This is a lightweight estimate of the minimax-regret objective of PAIRED [Dennis et al., 2020], with the hint-equipped agent standing in for the antagonist, and it separates environments at

		Agent / design BoxingGym	Chemistry & materials A-Lab, Coscientist	SLOWLAB (this work)
Slow	feedback latency parallel width	seconds–minutes unbounded	hours–days high (96/384-well)	210 days hard-capped, tiered
Noisy	signal-to-noise	deterministic	high	low (plant-to-plant)
Costly	cost of one trial cost of a retry	≈ 0 ≈ 0	reagents, instrument time low–moderate	a crop, priced in the objective one growing season
Irreversible	can a design be undone?	fully (reset)	largely (re-run)	no

Table 1: Evaluation environments against the four conditions of §3. Each row is one measurable property of the condition naming it. The closest prior benchmark differs on every row.

the learning frontier from those already mastered or intractable. A benchmark needs exactly this, since the alternative is to assert that one’s tasks are well-posed. We report it as a diagnostic rather than adopt it as our score: with four hand-built configurations we can price a task directly against what a perfect reasoner could extract from it (§3.3.2), whereas hint regret estimates reachability from a solver’s behaviour, which is what one needs when the environments are generated at scale. It earned its place by correcting us. SCREEN is the task we came closest to withdrawing, since its ceiling sits below the noise a full budget leaves (Table 2); it also carries the largest hint gap of the tasks the diagnostic scores, 0.27 against 0.15–0.17. A task can be at the frontier of what a solver can do and still be one where instance-tailoring is worth little, and only one of the two instruments sees each fact. Note that “validity” in that work means executability; the word is already claimed in this literature for something other than inferential soundness.

Self-driving laboratories. Chen et al. [2026] survey deployed self-driving laboratories and argue that evaluation should “prioritise decision quality under constraints and reproducibility rather than single-number performance in idealised settings”. They prescribe the reference agents we adopt—“random and space-filling designs, standard BO for short-horizon tasks”—with the aim of providing “stable reference points that reflect decision structure, not to maximise baseline performance” (Appendix H). What is missing is an executable environment in which any of it can be measured.

Adjacent benchmarks. ScienceAgentBench, DiscoveryBench and related suites [Chen et al., 2025, Majumder et al., 2024] evaluate analysis and reproduction of published findings, where the data already exist and the agent bears no collection cost. WOFOSTGym and CropGym expose crop simulators as RL *control* problems [Coyle et al., 2025]—the agent sets irrigation over time rather than committing to a design and waiting; we reuse the framing but not the task. A substantial literature builds autonomous laboratories [Szymanski et al., 2023, 2026, Burger et al., 2020, Boiko et al., 2023, M. Bran et al., 2024], surveyed by Stach et al. [2024] and Abolhasani and Kumacheva [2023]; those systems operate on hour-to-day timescales and report constraint-violation behaviour rather than releasing an executable benchmark.

3 SLOWLAB

3.1 Regime and problem formulation

SLOWLAB targets sequential experimental design under four conditions that hold jointly. **Slow**: what can be learnt per unit of calendar time is bounded by feedback latency *and* by a hard ceiling on how many trials run concurrently—a facility with a thousand chambers and a four-month cycle is not slow in the sense that binds. That ceiling is *tiered*, because some factors are set for a whole block of units and others per unit, so replication and coverage compete for the same physical resource. **Noisy**: a single unit does not settle a question, which is what turns replication, blocking and allocation into decisions rather than formalities. **Costly**: a trial consumes resources measured in the objective’s own units, so exploration is priced rather than free. **Irreversible**: a submitted design cannot be rolled back—no branch, no checkpoint, no revert—so the only remedy for a bad design is a new one in a later cycle, and the units it occupied are gone. Table 1 places existing evaluation environments against these conditions. We instantiate the regime in controlled-environment agriculture (§4.2).

Problem formulation. An instance is a parameter vector $\theta \sim P_\Theta$, unknown to the agent, determining a response $f_\theta : \mathcal{X} \rightarrow \mathbb{R}$ on a factor space $\mathcal{X} \subseteq \mathbb{R}^d$ and measured in the objective’s own units. Over R rounds the agent commits a *design* $D_r \in \mathcal{D}$ — a finite set of treatments in $\mathcal{X}$ together with their assignment to the physical trial slots $\mathcal{U}_r$ — and states a recommendation $\hat{x}_r \in \mathcal{X}$. The admissible set $\mathcal{D}$ belongs to the facility rather than to the task; §3.2.3 gives it for our instantiation. Its one structural role here is that it is not all finite multisets: an agent cannot buy precision and coverage with the same slots.

Committing holds $\mathcal{U}_r$ for a cycle of length Δ , after which each slot returns one observation

$$y_u = f_\theta(x_u; \phi_u) + \varepsilon_u, \quad \varepsilon \sim \mathcal{N}(0, \Sigma(D_r)), \quad (1)$$

where x_u is the treatment assigned to slot u and ϕ_u an independent per-slot draw of the response model’s own parameters. Two features of (1) carry **noisy**. Drawing ϕ per slot rather than adding a scalar error makes the variance heteroscedastic in x and structured by the mechanism; and $\Sigma(D_r)$ is not diagonal, so *where* a treatment is placed changes what the round can resolve, not only which treatments were chosen. Appendix F gives Σ for our instantiation.

The other three conditions restrict the action. Writing $\mathcal{F}_r = \sigma(\{y_u : u \in \mathcal{U}_1 \cup \dots \cup \mathcal{U}_{r-1}\})$ for everything returned before round r , and $\mathcal{A}_r$ for the entire action set available at round r ,

$$D_r, \hat{x}_r \text{ are } \mathcal{F}_r\text{-measurable,} \quad \mathcal{A}_r = \{(D_r, \hat{x}_r)\}. \quad (2)$$

The first is **slow** stated as measurability rather than as sensor deprivation: information about a committed slot may well exist before its cycle ends, but no action is available on it, and Appendix L names the two actions that would separate the two readings. The second is **irreversible**: the action set contains commit and recommend and nothing else, from which it follows that no operation shrinks $\mathcal{F}_r$, releases a slot early, or refunds one already committed. **Costly** is carried not by a constraint but by the objective, which prices a trial in the units it is trying to maximise, so that a campaign has a cost beside the quality of its answer (§3.3).

The problem. Given $\mathcal{F}_r$ and nothing else, the agent chooses $D_1, \dots, D_R$ and finally commits to $\hat{x}_R$. It is judged on how far that recommendation falls short of the instance’s own optimum,

$$\mathcal{R} = f_\theta(x_\theta^*) - f_\theta(\hat{x}_R), \quad x_\theta^* \in \arg \max_{x \in \mathcal{X}} f_\theta(x), \quad (3)$$

and, separately, on what the campaign spent reaching it: every slot it committed is charged the same shortfall, at the treatment that slot was given, for a full cycle (5). Neither quantity is observable to the agent — θ , f_θ and x_θ^* are all withheld — and neither can be estimated from $\mathcal{F}_r$ without the design that generated it.

The two pull against each other, and the constraints are what stop the usual resolution. Reducing (3) calls for slots spent on treatments the agent expects to be poor, since those are the ones that discriminate; that is exactly what the second charge penalises. In an environment where a trial is a function call, the tension is nominal: try something, look, retract it. Here (2) removes retraction, (1) means one slot does not settle whether a treatment was poor, and $\mathcal{D}$ means the slots spent discriminating are slots not spent confirming. A campaign therefore occupies the facility for $R\Delta$ and consists of R ordered, unrepeatable commitments, each made before the information that would have justified it exists. §3.3 states both quantities precisely and what they are read against; Appendix B works one round through in full.

3.2 The environment

SLOWLAB is a harness: the components below are defined over a domain rather than within one, and §4.2 supplies the first domain that fills them in; Figure 1 on page 3 is the loop this subsection specifies.

3.2.1 Interaction protocol

Figure 2 is the whole interface. `validate_design` is free and unlimited; `submit_design` commits the named slots for one cycle; `advance` runs the cycle and returns one value per slot; the interim recommendation records what the agent would advise if the campaign stopped there, which supplies the regret trajectory. Budget is accounted in slot-cycles. The recommendation is elicited, never consumed: it costs no budget,

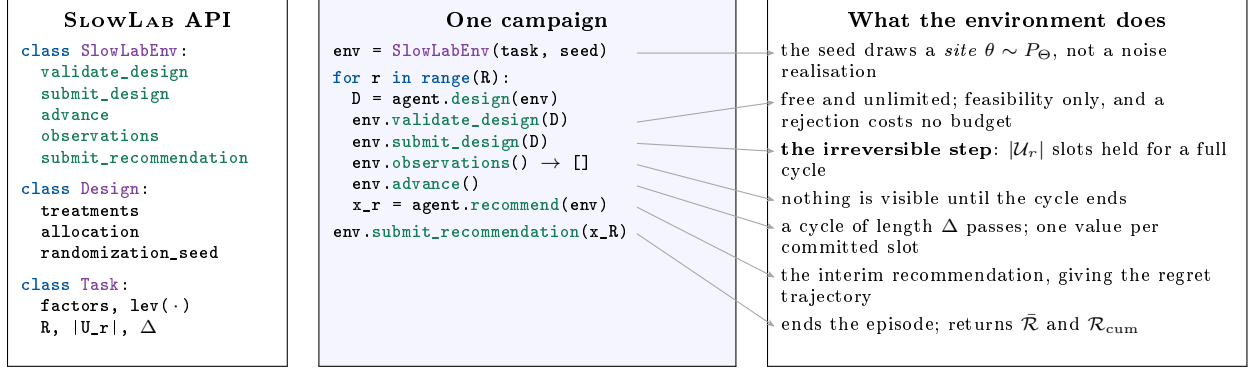


Figure 2: **The interaction protocol.** (*left*) What an agent touches: an environment exposing five calls, a **Design** naming treatments and assigning them to named slots, and a **Task** fixing the factors, their control levels, the number of rounds, the width cap and the cycle length. (*centre*) One campaign. (*right*) What each call costs. Checking is free and a rejection carries a code and a detail, so an agent may revise without penalty; committing is the step that cannot be undone. Between `submit_design` and `advance` the observation list stays empty — a regression test guards it — which is (2) in code.

returns nothing, and enters no later design — an agent that revises it every round is not penalised for having been wrong earlier. Design and recommendation are separate submissions, which is what allows an agent to run a good campaign and then answer badly, and §3.3.2 to score the two apart.

3.2.2 Action space

A submission is a **Design**: named treatments, each a full vector of factor values; an allocation mapping treatments to specific units; and a declared randomisation seed. The agent chooses *what* to test, *how many units* each treatment receives, and *which* units those are—replication and allocation are decisions rather than defaults. It does not choose what to measure, when to measure, or whether to intervene mid-cycle; a submitted design runs to completion.

3.2.3 Facility and control levels

A facility is C blocks of L units. Every factor carries a *control level*: a BLOCK-level factor takes one value for a whole block, a UNIT-level factor is set per unit. The level is a physical property of the apparatus, not a modelling choice, and a design assigning two different block-level values inside one block describes an experiment that cannot be performed.

Two consequences follow, and we learned both by implementation. With k block-level factors, the number of distinct block-level *combinations*—not levels per factor—bounds the number of whole-plot treatments by C . And if a combination occupies exactly one block, its effect is confounded with that block’s, and unit-level replicates cannot be spread. A valid design is therefore a **split-plot**: block-level factors as whole plots replicated across at least two blocks, unit-level factors as subplots. This structure is what makes allocation a decision, and it is absent from any environment in which an experiment is a function call.

3.2.4 Forward model and instances

The environment takes a published forward model f_{θ} mapping a factor vector to a mean response, and a parameter distribution P_{Θ} describing how that model’s coefficients vary across sites in the literature. A seed selects an *instance* by drawing $\theta \sim P_{\Theta}$: a different problem, not the same problem observed again. The response surface and the location of its optimum are therefore fixed by the published model rather than by us.

Within an instance, each unit receives its own draw of the model’s parameters rather than an additive noise term. Unit-to-unit variation *is* parameter variation, and modelling it this way yields heteroscedastic,

temporally correlated variability without further assumptions. The nuisance effects of §3.1 are what an allocation must avoid confounding with.

3.2.5 Objective and cost

The objective is a rate: revenue per unit area per unit time, less the full cost of the resources the treatment consumed, $f_\theta(x) = (\text{revenue}(x) - \text{cost}(x))/\Delta$. Pricing is not decoration. Under a pure-yield objective a forward model is typically monotone in its resource inputs, so their optima sit on the boundary and those dimensions are trivial; charging for them is what produces interior optima. Expressing the cost in the objective’s own units is also what makes the *Costly* measure interpretable: a campaign’s expenditure is directly comparable to what it was trying to maximise.

The costs enter at weight one, and this is enforced rather than chosen. An earlier version carried a multiplier λ on the cost term, declared as a task-design parameter. It was not one: it set where every optimum sat, and at $\lambda = 0.25$ it pushed four of six factors onto their rails. The environment now raises if $\lambda \neq 1$ (Appendix L). What TRANSFER moves mid-episode is a price in the cost model, not a weight on it, so the objective stays the profit a grower would actually bank in both halves of that task.

3.2.6 Feasibility checking

`validate_design` checks *mechanical* feasibility only—control levels, unit conflicts, budget, factor ranges—and is free and unlimited; rejections cost no budget but are counted. The design measure of §3.3.2 is computed at scoring time and cannot be called by the agent. Separating the two has a purpose: because feasibility is free to check, a design that passes it yet is scientifically poor is unambiguously a reasoning error rather than an interface error. Anything requiring a statistical assumption—assumed variances, power targets—belongs to the tool layer, an experimental condition to be ablated (Appendix H), not a free service of the environment.

3.2.7 Language-model interface

A language model reaches the same interface through a harness that serialises the round. It receives the facility layout and unit naming, the budget and rounds remaining, each factor’s range and control level, the hierarchy constraint, every observation so far with the settings that produced it, the currently available units, and a JSON schema; it returns treatments, an allocation, a randomisation seed, and the recommendation it would give if the campaign stopped now, which supplies the trajectory in (4).

The prompt states constraints and never gives advice: the words replication, randomisation, blocking, split-plot, pure error and confounding do not appear, since those name the capability under test, and a regression test enforces their absence. Failures are counted separately—a reply that will not parse from a design that is mechanically infeasible—both fed back verbatim and retried up to three times, and neither consuming budget, consistent with `validate_design` being free. “Cannot emit valid JSON” and “cannot design an experiment” are therefore different numbers. Exhausting the retries forfeits that round and the campaign continues, so one unparseable reply is not as costly as a failed campaign.

3.2.8 Configuration and extension

Configurations are declarative: factors and their control levels, facility shape, rounds, per-round width, variance components, cost weight. Adding one requires no environment code. Agents implement a single `run(env)` method and are registered by name; the reference strategies of Appendix H and the harness above use the same interface, and the harness needs only a callable mapping a message list to a string, so no vendor SDK is required. A new domain supplies f_θ , P_Θ , a cost function in the objective’s units, and a facility whose units are not interchangeable (Appendix A).

3.3 Evaluation

Two quantities are reported per episode and never combined. *Realised regret* is read from what the campaign did; the *regret a perfect reasoner would still carry* is read from what each design could have supported, before its outcome is seen. Neither elicits anything from the agent beyond a design and a recommendation.

3.3.1 Realised regret

Both quantities of §3.1 are made precise here. They are regret against the same optimum in the same currency, because the experiment is paid for in the units of what it is trying to maximise. The first is (3), elicited after *every* round rather than only at R , and averaged over instances:

$$\bar{\mathcal{R}}(r) = \mathbb{E}_{\theta \sim P_{\Theta}} \left[f_{\theta}(x_{\theta}^*) - f_{\theta}(\hat{x}_r) \right], \quad x_{\theta}^* \in \arg \max_{x \in \mathcal{X}} f_{\theta}(x). \quad (4)$$

The second is what the experiments themselves forwent: every occupied slot is run at the treatment it was given, so a slot assigned a poor one costs the gap to the optimum for a full cycle of length Δ ,

$$\mathcal{R}_{\text{cum}} = \Delta \sum_{r=1}^R \sum_{u \in \mathcal{U}_r} \left[f_{\theta}(x_{\theta}^*) - f_{\theta}(x_u) \right]. \quad (5)$$

This is the usual simple-versus-cumulative pair, and the second term is usually dropped in benchmark settings because a trial there costs nothing. Here an agent that locates the optimum by first running every bad treatment has answered the question and exhausted the facility that paid for it. Neither implies the other, and in our runs they do order agents differently (Appendix H); an agent that runs no experiments forgoes nothing and is judged by (4) alone.

3.3.2 Design and decision quality

A campaign can fail in two ways that a single score cannot tell apart: the experiments could not have answered the question, or they could and the agent did not read them. Both are measured against $\mathcal{R}^*(D_r)$, the regret still left after a decision-maker who reasoned perfectly from this design’s outcome,

$$\mathcal{R}^*(D_r) = \underbrace{\mathbb{E}_{p(y|D_r)}}_{\text{over outcomes not yet seen}} \left[\underbrace{\min_{x \in \mathcal{X}}}_{\text{best recommendation}} \underbrace{\mathbb{E}_{p(\theta|y, D_r)} [f_{\theta}(x_{\theta}^*) - f_{\theta}(x)]}_{\text{its expected regret}} \right]. \quad (6)$$

Read it from the inside out. The bracket is the regret (3) of recommending x at site θ , averaged over the posterior a perfect reasoner would hold after seeing y , the vector of observations (1) that design D_r returns. The min picks the recommendation that posterior supports best. The outer expectation averages over outcomes the design has not produced yet, so $\mathcal{R}^*$ is a property of the design alone and can be computed before it is run. No agent can compute it: doing so needs P_{Θ} and the forward model, which the interface withholds. Written $\mathcal{R}^*(\emptyset)$ for a design that runs nothing, it is the regret of the best single recommendation, and so the most any experiment on the task could remove.

From it we form one ratio for what was run and one for what was concluded,

$$\underbrace{c(D_r) = \frac{\mathcal{R}^*(\emptyset) - \mathcal{R}^*(D_r)}{\mathcal{R}^*(\emptyset)}}_{\text{design efficiency}}, \quad \underbrace{\eta(D_r) = \frac{\mathcal{R}^*(D_r)}{\bar{\mathcal{R}}(r)}}_{\text{decision efficiency}}. \quad (7)$$

Design efficiency c is the fraction of the available risk $\mathcal{R}^*(\emptyset)$ that the experiments removed; it is 0% for a design that teaches nothing and 100% for one that identifies the optimum exactly. *Decision efficiency* η divides what the design made available, $\mathcal{R}^*(D_r)$, by what the agent actually achieved with it, $\bar{\mathcal{R}}(r)$ from (4); it is near 100% when the recommendation is as good as the data allowed and near 0% when the agent left most of it unused. The *decision* here is the recommendation $\hat{x}_r$ and nothing else: both terms of η are the regret of a point in $\mathcal{X}$, against the same optimum, and differ only in who chose it — the min inside (6), or the agent. Both are reported as percentages, larger is better for both, and an agent can fail at either.

Neither is new. $\mathcal{R}^*(\emptyset)$ and $\mathcal{R}^*(D)$ are the prior and preposterior Bayes risk of the decision problem, so their difference is the expected value of sample information [Raiffa and Schlaifer, 1961, Howard, 1966], equivalently a Knowledge Gradient [Frazier et al., 2008] for a batch rather than a single point. What is ours is the pairing. Unlike information gain about θ [Gandhi et al., 2025], (6) scores the recommendation rather than the parameters, so a design that pins θ down while spending its budget far from the optimum scores low; and since (6) contains no $\hat{x}_r$, c is a functional of the design and the instance distribution alone while η carries $\bar{\mathcal{R}}(r)$ in its denominator. The two cannot be restatements of each other, and §4.4 shows one intervention moving them in opposite directions.

Estimation. We discretise P_Θ into M atoms and score each round’s design against the prior rather than against the realised history, because conditioning on a history collapses the effective atom count below five by the third round. The obvious estimator — the same atoms forming the posterior, choosing the recommendation and grading it — is in-sample selection and is badly optimistic: it reports $c = 97\%$ for a design worth -64% on held-out atoms. We temper the likelihood, choose the temperature by leave-one-out over the inference atoms, and grade on a disjoint half. Appendix F gives the mechanics, the diagnostics, and what each change is worth. What survives estimation is an ordering, so we report c and η as ranks and as means over dozens of designs, and draw conclusions only from arms compared pairwise at fixed M ; Appendix F says what forces that.

4 Experiments

4.1 Experimental setup

We run five language models — DeepSeek-v4-flash, GPT-5.6-luna, Qwen3.8-27B, MiMo-v2.5 and GLM-5.3-flash — on all four tasks, twice each: once with the interface of §3.2.1 alone and once with two standard tools added (§4.4). That is 40 model-task cells and 800 episodes. Four scripted strategies are run on the same instances and reported in Appendix H. All five models are run at one sampling temperature under one prompt and come from a single price tier, which is enough to show that a habit is not one model’s quirk and not enough to characterise the field; the prompt is the variable we would vary first, since we fixed one defect in it mid-study that cut infeasible submissions roughly fivefold.

Every task is run on the same 20 instances. A seed draws a site, a parameter vector $\theta \sim P_\Theta$ with its own optimum, rather than a noise realisation, so all agents face identical problems and every contrast below is paired by site. An agent is a single `run(env)` call with access to nothing beyond the interface of §3.2.1: it is not told P_Θ , the forward model, or that a reference exists, and regret is computed afterwards by the evaluator from x_θ^* , which no agent can read. Within a task all agents share the facility, budget, factor ranges and cycle length; only the strategy varies.

Feasibility checking is free and unlimited for every agent and a rejected design costs no budget (§3.2.6). A scripted reference uses that freedom when it is written, by never proposing a design the facility cannot run; a language model uses it at run time, through the retry loop of §3.2.7.

4.2 Domain: agriculture

Several domains satisfy the four conditions: microbial evolution, bioprocess scale-up, materials ageing, dose-finding trials. SlowLab is built so that adding one requires no environment code (§3.2.8). We build the first in controlled-environment horticulture because the conditions there hold jointly and by nature: a crop occupies its greenhouse for the better part of a year, plant-to-plant variation is large, running a treatment means growing and selling a crop, and heat stress destroys fruit that cannot be recovered. None of this is a dial we set, and a benchmark whose delay, noise and cost were independent hyperparameters could be tuned until the desired result appeared. Horticulture also supplies the two things a domain must bring: a mechanistically modelled response with four decades of published parameters, and a priced output, so that both f_θ and the cost of experimenting come from the literature rather than from us.

Appendix G gives the coefficients with their sources, and Appendix L names the four that are ours rather than the literature’s and what each is worth. Filling in §3: f_θ is the reduced state-variable TOMGRO model [Jones et al., 1999] and P_Θ its published between-site parameter spread, so a seed draws a real site rather than a perturbation we chose. Blocks are climate chambers and units are independently plumbed hydroponic loops, which is what puts four of the six factors at chamber level and the other two at loop level. The objective is net profit per square metre per day, priced with public European coefficients. The same appendix explains the choice of a mechanistic rather than a Gaussian-process ground truth and the caveats that follow from it.

4.3 Tasks

The four are the stages an industrial experimental campaign goes through, and each isolates a capability the previous one does not require. They differ in the question asked and in the geometry that makes it hard, never in which of the four conditions applies. The variance components, the cycle length, the per-unit cost

Task	What the agent must do	Factors	Facility	$R \times$ units	days	$\mathcal{R}^*(\emptyset)$	$\sigma_{\text{tot}}/\sqrt{B}$	ratio
SANITY	Close the loop at all	1 (day temp.)	4×4	2×8	420	0.0055	0.0041	1.3
SCREEN	Find which factors matter	6 (management)	8×3	2×12	420	0.0044	0.0058	0.8
OPTIMISE	Find the optimum, respecting the facility	2 (mixed level)	4×3	3×12	630	0.0061	0.0028	2.2
TRANSFER	Survive an energy price shock	4 (resource)	8×3	3×24	630	0.0041	0.0042	1.0

Table 2: The four tasks. *Facility* is chambers $\times$ loops and $R \times$ units is rounds times units per round; *mixed level* means one chamber-level and one loop-level factor. $\mathcal{R}^*(\emptyset)$ is (6) with no data: the most any experiment on the task could remove. $\sigma_{\text{tot}}/\sqrt{B}$ is the standard error of the mean response a full budget of B units achieves (Appendix C). *ratio* is the first over the second.

and the commitment structure are identical across all four; only the question and the geometry change.

SANITY is the control. One factor and sixteen unit-slots make it easy, but its variance components are those of the other three, since relaxing a condition on one task would contradict the claim that we do not relax them. Any competent closed loop should beat random search here, so a method that does not has an interface problem rather than a reasoning problem.

SCREEN tests allocation under scarcity. It varies all six management factors the domain exposes on a budget covering one unreplicated sweep, which forces a choice between precision and coverage, and two rounds leave no room to iterate out of it. Effect sparsity is not injected: it follows from crop physiology and the price list, and one factor is rail-bound because at 2024 European electricity prices supplemental light never repays its energy, so recognising that is part of screening.

OPTIMISE tests iteration under a control hierarchy. One chamber-level and one loop-level factor mean a valid design must be a split-plot, so where an agent puts a treatment matters as much as which treatment it is.

TRANSFER tests whether an agent modelled the mechanism or the surface. Four resource factors are optimised at the site’s own prices; electricity and gas then both rise by a factor of 2.2 and the agent must recommend again *with no further experiments*. The shock size is the 2021–2022 European energy crisis, over which Eurostat’s non-household gas price reached an all-time high of 0.0867 €/kWh and non-household electricity excluding taxes peaked at 0.1987 €/kWh, both two to three times their pre-crisis level [Eurostat, 2025]. It falls only on energy: the tomato price, transplants, labour and purchased CO₂ do not move. Every observation returns a revenue rate, an energy cost rate and an other-cost rate alongside their combination, to the scripted and language-model interfaces alike, so an agent that modelled the three separately can re-solve without running anything and one that modelled only the margin cannot.

How much each task can reward. The last two columns of Table 2 are the ceiling $\mathcal{R}^*(\emptyset)$ and its ratio to the standard error of the mean response that a full budget can achieve. Only OPTIMISE is comfortable, at 2.2. SANITY sits at 1.3, and SCREEN and TRANSFER at 0.8 and 1.0: on those two the whole of what instance-tailoring could win is at or below what the noise a full budget leaves behind. They are the tasks on which every agent in §4.4, scripted and otherwise, converts a few per cent of what its design carried — the two facts are the same fact — and, at twenty instances, the tasks whose five-model ordering does not survive resampling the sites: split-half rank correlation runs in the same order as this column, from -0.28 on SCREEN to $+0.76$ on OPTIMISE (Appendix J). What each task supports is therefore different, and §4.4 reads the two low-ratio tasks only through proportions of submitted designs and through within-model paired contrasts, never through a leaderboard position. The cause is not noise we chose. It is that exposing more factors drives the optimum onto constraints where the remaining factors are worth little to locate, which Appendix L traces. We report the ratio and read the results in its light; we stopped using it to admit or exclude tasks, because that makes the task set a function of the simulator’s fine detail.

Four scripted strategies — random search, a Plackett–Burman screen followed by a response surface, and Bayesian optimisation at one and two replicates per treatment — are released with the environment and run on all four tasks in Appendix H. They are there so that a reader can see what standard methods do here. They are not the comparison this paper draws: the ceiling $\mathcal{R}^*(\emptyset)$ is, because it does not depend on anyone’s implementation choices.

Model	0-shot	$\bar{\mathcal{R}} \downarrow$	$\mathcal{R}_{\text{cum}} \downarrow$	$c \uparrow$	$\eta \uparrow$
SANITY $\mathcal{R}^*(\emptyset) = 0.0055$					
DeepSeek-v4-flash	0.0375	0.0279(55)	258	6%	18%
GPT-5.6-luna	0.0269	0.0131(26)	297	12%	37%
Qwen3.8-27B	0.0256	0.0419(135)	239	8%	12%
MiMo-v2.5	0.0227	0.0265(65)	252	11%	19%
GLM-5.3-flash	0.0182	0.0202(72)	297	8%	25%
SCREEN $\mathcal{R}^*(\emptyset) = 0.0044$					
DeepSeek-v4-flash	0.2444	0.1037(195)	943	5%	4%
GPT-5.6-luna	0.1628	0.0803(93)	845	6%	5%
Qwen3.8-27B	0.2909	0.0893(121)	814	6%	5%
MiMo-v2.5	0.1992	0.0871(86)	603	7%	5%
GLM-5.3-flash	0.1865	0.0866(84)	717	7%	5%
OPTIMISE $\mathcal{R}^*(\emptyset) = 0.0061$					
DeepSeek-v4-flash	0.0412	0.0365(41)	405	6%	16%
GPT-5.6-luna	0.0248	0.0110(21)	450	18%	45%
Qwen3.8-27B	0.0288	0.0291(63)	383	8%	19%
MiMo-v2.5	0.0259	0.0267(51)	508	7%	21%
GLM-5.3-flash	0.0245	0.0115(20)	437	14%	46%
TRANSFER $\mathcal{R}^*(\emptyset) = 0.0041$					
DeepSeek-v4-flash	0.2300	0.0758(100)	1896	7%	5%
GPT-5.6-luna	0.1635	0.0544(45)	1896	4%	7%
Qwen3.8-27B	0.2427	0.0711(132)	1822	9%	5%
MiMo-v2.5	0.2102	0.0778(133)	1976	9%	5%
GLM-5.3-flash	0.2022	0.0615(101)	2004	4%	6%

Table 3: The four quantities of §3.3, for language models without tools, 20 instances per cell. *0-shot* is the model’s own recommendation before it sees any data, and is not one of the four. $\bar{\mathcal{R}}$ (4) is realised regret after the full campaign, standard error in the last two places, in bold where it beats that model’s own 0-shot; $\mathcal{R}_{\text{cum}}$ (5) is what the experiments themselves forwent, in the same currency summed over occupied slot-days; c and η (7) are design and decision efficiency under the cross-validated estimator of §3.3.2, to be read as ranks. η near 100% would mean the recommendation is as good as the design allowed. TRANSFER is scored at the training prices here so the four tasks are comparable; the post-shock number the task exists for is in §4.4. No cell reaches $\mathcal{R}^*(\emptyset)$.

4.4 Results

Five models, four tasks, 20 instances each, run twice: once with the interface of §3.2.7 alone and once with two standard tools added — a screening design generator, and a posterior-maximisation routine fitted to whatever observations the agent has collected. That is forty model–task cells, every one with its full twenty instances. We report the two halves of §3.3 in turn, realised regret first and design and decision quality second, then the two effects that only appear because the benchmark elicits a recommendation every round and runs a paired tool arm.

4.4.1 The benchmark is far from saturated

No arm reaches the ceiling. $\mathcal{R}^*(\emptyset)$ is the regret a perfect reasoner carries having run *no* experiment at all. Not one of the forty arms reaches it. The bare arms sit between $1.8\times$ and $23.4\times$ above it, widest on the two tasks with the most factors — $18\text{--}23\times$ on SCREEN and $13\text{--}19\times$ on TRANSFER against $2\text{--}8\times$ on the two smaller ones — and the scripted references of Appendix H are closer but also short, at $1.6\times$ to $22.3\times$. Against the weaker reference of a model’s own zero-shot recommendation, thirty-one of the forty cells end below it and nine end *above*: two or three growing cycles left the model advising something worse than it would have advised for free. All nine are on SANITY and OPTIMISE, where the zero-shot recommendation is already close enough that noisy data can only move it the wrong way. The benchmark is not saturated, and

Model	SANITY needs $k \geq 2$		SCREEN needs $k \geq 7$		OPTIMISE needs $k \geq 3$		TRANSFER needs $k \geq 5$	
	<i>bare</i>	<i>+tools</i>	<i>bare</i>	<i>+tools</i>	<i>bare</i>	<i>+tools</i>	<i>bare</i>	<i>+tools</i>
(a) % of designs that cannot separate the main effects (rank-deficient)								
DeepSeek-v4-flash	4	10	100	66	57	54	96	66
GPT-5.6-luna	0	0	100	56	9	20	94	71
Qwen3.8-27B	9	6	96	47	51	44	81	54
MiMo-v2.5	0	0	100	57	90	57	95	65
GLM-5.3-flash	0	6	100	69	64	59	95	66
(b) % of designs with fewer than $d + 1$ distinct treatments — why (a) happens								
DeepSeek-v4-flash	4	10	100	62	57	54	86	66
GPT-5.6-luna	0	0	100	56	5	12	92	71
Qwen3.8-27B	9	6	89	45	41	44	68	52
MiMo-v2.5	0	0	100	57	89	54	95	63
GLM-5.3-flash	0	6	100	69	59	56	93	63

Table 4: What the submitted designs look like. Every cell is a percentage of that model’s designs on that task, pooled over every round of all 20 instances; *bare* is the interface alone and *+tools* the same model with the two tools. **(a)** A design is rank-deficient when its distinct treatments, stacked beside an intercept column, do not reach rank $d + 1$, so the d main effects cannot be estimated separately. **(b)** The immediate cause, and almost the whole of it: k distinct treatments give rank at most k , so a design needs $k \geq d + 1$ before replication can buy anything. The threshold is the task’s, not a fixed number — SANITY needs two and SCREEN seven — and (b) tracks (a) to within a few points on every task. The screening tool repairs much of (a) on the two higher-dimensional tasks and none of it on SANITY, which has one factor and nothing to screen.

it is least saturated where the design problem is hardest. $\mathcal{R}^*(\emptyset)$ is a ceiling rather than a target: no design we can find removes more than a third of it in one round (Appendix I), which is why the gap to it is read as a scale and the comparisons that follow are made between arms.

The two regrets order agents differently. Realised regret and cumulative regret are not substitutes, which is why §3.3.1 reports both: on SANITY and OPTIMISE, the two tasks whose orderings are stable enough to read (Appendix J), they rank-correlate at -0.80 and -0.60 within task, so the model with the best final answer is among the most expensive campaigns to have got there. An agent judged on the recommendation alone is free to burn the facility reaching it.

TRANSFER costs more after the shock than before it. The column in Table 3 is priced at the training energy costs so the four tasks are comparable. The quantity the task exists for is the one after: electricity and gas rise by 2.2 and the agent must re-solve with no further experiments. Every arm is worse there, by factors of 1.06 to 1.98, and each tooled arm is worse than its bare twin. Two reference agents isolate what is being tested — identical in algorithm, budget and observations, differing only in whether they modelled gross margin or modelled revenue and the two cost streams separately, they score identically before the shock and differ by $1.47\times$ after it (0.0605 against 0.0411, paired $t = 5.5$ over 40 sites).

4.4.2 Designs and decisions fail independently

A campaign can fail because the experiments could not have answered the question, or because they could and the agent did not read them. (7) separates the two, and both are failing here.

The designs could not have answered the question. Unaided, 99% of designs on SCREEN, 92% on TRANSFER and 54% on OPTIMISE are rank-deficient over the factors the task varies, so the main effects they were commissioned to separate are not estimable at all (Table 4a, the *bare* columns). Panel b gives the cause

and nearly all of it: 98%, 87% and 50% of those designs carry fewer distinct treatments than the task has factors plus one. The habit behind the number is the same everywhere — the median design uses *two* distinct treatments replicated across the whole facility, whether the task needs two or seven. These are not careless designs. All but four of the 1137 submitted declare a randomisation seed, each distinct treatment carries a median of three to twelve units depending on the task, and the allocation respects the control hierarchy; they are designs that buy precision with the units coverage needed, and under a hard parallelism cap the two draw on the same physical units. An agent that always buys precision is under-powered against its own question, which is the opposite of the failure we built the benchmark expecting. Appendix I separates this from the alternative reading, that a low c simply means the task is hard: searching the admissible set with knowledge of P_Θ and the forward model finds designs reaching c of 27% to 37%, so the best any agent reaches here is $1.4\times$ to $2.8\times$ short of what one round can buy. The same search puts the median narrow design well below the median wide one on every task, which is the form the width diagnosis actually takes: a well-placed pair of treatments can do well, and a typical one does not.

The decisions then use a small fraction of what those designs allowed. Decision efficiency is the half of (7) that does not blame the design: it divides the regret a perfect reasoner would carry *on the design the agent actually submitted* by the regret the agent carried. The η column of Table 3 runs from 4% to 46% over the twenty bare cells, and splits by task rather than by model — 22% on SANITY and 29% on OPTIMISE against 5% on SCREEN and 6% on TRANSFER. On the two higher-dimensional tasks the campaigns leave roughly nineteen twentieths of what they bought on the table.

The two failures are not the same failure. Within task, c and η rank-correlate at $+0.80$ on OPTIMISE, which is the one task whose five-model ordering survives resampling the twenty sites (Appendix J); the two orderings come apart on TRANSFER as well, but there neither is reproducible and we do not read it. A single score would have reported only their product, and §4.4.3 shows an intervention that moves them in opposite directions within a model and paired by instance, which is the evidence that does not depend on any ordering at all.

4.4.3 Standard tools improve the design and damage the answer

This is the sharpest evidence we have that the two halves of (7) measure different things, because it is one intervention applied within a model and paired by instance rather than a correlation across agents — two numbers that moved together would be one number. The screening tool works: rank-deficiency falls from 99% to 59% on SCREEN and 92% to 64% on TRANSFER, and c rises in seven of those ten cells by up to 6.5 points, the other three moving by -0.5 to -0.2 , inside the estimator’s noise. In the same ten cells η falls and realised regret rises, by 4% to 59%; over all four tasks the tools make 16 of 20 pairs worse and four better, six significant at paired $p < 0.05$ — four worse and two better, both on OPTIMISE (Table 5).

The mechanism is that the inference tool inherits whatever the design can support. It is the same routine our Bayesian-optimisation reference uses for its own recommendation, and that reference carries 0.0172, 0.0586, 0.0120 and 0.0335 on the four tasks against 0.0711, 0.1257, 0.0219 and 0.0901 for the identical routine fitted to a language model’s design. Nothing about the routine changed, only what it was fitted to, and what it was fitted to is thinner: on the three multi-factor tasks the reference’s campaigns hand it 1.8 to 3.7 times as many distinct points. Full rank is not sufficiency: the median tooled design on SCREEN is thirteen points over six factors, barely one level a factor, and its main effects are estimable long before its surface is. Appendix H rules out the obvious alternative, that our tool is simply misconfigured. The models then absorb roughly a third of the tool’s disadvantage: regressing the tools’ effect on how much worse the tool’s own proposal is than the model’s gives a slope of 0.29 ± 0.09 at $r = 0.62$, $p = 0.004$, with the sign predicted in eighteen of twenty cells. A third is an average over cells, and it hides how literal the deference becomes round by round: in 384 of the 693 turns where both are visible — 55%, and between 53% and 62% on every task — the recommendation the model states is the tool’s proposed point to six digits. The models are neither credulous nor immune, and none of them says the tool is wrong — which is the regime’s doing rather than the tool’s. Detecting a bad tool is ordinarily cheap: run its recommendation, see the number, stop using it. Here that costs a growing cycle and the agent has at most three, so what the benchmark measures is not whether a model can use a tool but whether it can price one it cannot afford to test.

Task	Model	Δc design	$\Delta \eta$ decision	$\Delta \bar{\mathcal{R}}$ final answer	paired t
SANITY					
	DeepSeek-v4-flash	-1.1	-5.3	+42%	+1.5
	GPT-5.6-luna	-4.9	-11.5	+53%	+1.0
	Qwen3.8-27B	-1.3	+3.6	-22%	-0.6
	MiMo-v2.5	-4.8	+3.0	-9%	-0.4
	GLM-5.3-flash	-3.3	-15.5	+170%	+2.2*
SCREEN					
	DeepSeek-v4-flash	+5.4	-1.2	+32%	+1.8
	GPT-5.6-luna	+5.2	-1.8	+45%	+3.3*
	Qwen3.8-27B	+5.9	-1.1	+24%	+1.6
	MiMo-v2.5	+2.1	-1.4	+39%	+2.9*
	GLM-5.3-flash	-0.3	-1.0	+25%	+1.7
OPTIMISE					
	DeepSeek-v4-flash	+0.7	+19.8	-56%	-4.1*
	GPT-5.6-luna	-7.1	-13.2	+53%	+1.3
	Qwen3.8-27B	-0.0	-5.5	+40%	+1.8
	MiMo-v2.5	-1.7	+17.8	-45%	-2.2*
	GLM-5.3-flash	-2.7	-15.5	+56%	+1.1
TRANSFER					
	DeepSeek-v4-flash	+1.8	-1.1	+26%	+1.2
	GPT-5.6-luna	+6.5	-3.0	+59%	+4.5*
	Qwen3.8-27B	-0.2	-0.7	+16%	+1.0
	MiMo-v2.5	-0.5	-0.6	+16%	+0.9
	GLM-5.3-flash	+4.6	-0.5	+4%	+0.2

Table 5: What the two tools do, paired by instance, on three of the four quantities of Table 3. Δc is the change in design efficiency in percentage points, $\Delta \eta$ the change in decision efficiency also in percentage points, and $\Delta \bar{\mathcal{R}}$ the relative change in the regret of the final recommendation. A positive Δc with a positive $\Delta \bar{\mathcal{R}}$ is a better design with a worse answer, which is what most cells show. * marks $p < 0.05$ on a paired t -test over instances: six cells, four of them a worse answer and two a better one, both on OPTIMISE.

4.4.4 The first cycle of data often makes the recommendation worse

Eliciting a recommendation after every round rather than once at the end exposes something the endpoint hides. In 44% of episodes the recommendation after the first round is worse than the one the same model made with no data at all, and in 24 of the 40 cells the mean is worse. The split is by task: 68% of OPTIMISE episodes and 54% of SANITY, against 29% and 26% on SCREEN and TRANSFER, where the zero-shot recommendation starts far enough from the optimum that one round of almost any data helps. Slow feedback costs an agent in proportion to how many cycles it needs, and here the first cycle is often spent undoing itself.

5 Conclusion and future work

SLOWLAB asks an agent to do what a grower’s agronomist does: design an experiment, commit it, wait out a season, and design again. The four conditions are not scored, because in this domain they cannot be switched off; what is scored is the experimentation carried out while they hold, read both as the regret it realised and as the regret a perfect reasoner would still have carried given the same designs. One component does have a dial, and turning it sharpened the claim rather than confirming it: withdrawing a quarter of the facility after heat damage changes no agent’s realised regret, scripted or otherwise, so irreversibility here binds through commitment and not through capacity (Appendix L).

Run on five models, the instrument says what a single outcome number could not. No arm reaches what the instance distribution already allows with no data at all. The designs are not sloppy — they

replicate, declare their randomisation and respect the control hierarchy — they are too small, most using two distinct treatments whether the problem has one factor or six. And handing the models the method they failed to reconstruct moves the two halves of the score in opposite directions, within a model and paired by instance: the screening tool repairs the designs and the campaigns end worse, because the inference routine is fitted to those designs and inherits what they cannot support. Design quality and decision quality are two measurements, and that is the evidence.

Future work. The measure we would most like to add is a design score conditioned on the inference procedure that will actually read the design: (6) prices a design by what a Bayes-optimal reader could extract, and §4.4 shows the gap between that and what the agent’s own reader extracts is where the action is. The cheapest missing experiment is a second tooled arm whose inference routine is known to be good on these tasks, which would separate deference under slow feedback from tool use in general. We release the environment, the reference points, the tasks and the harness; Appendix L carries the register of what is still wrong with all four, Appendix F what the estimator cannot yet do, and Appendix A four task extensions the crop model already supports.

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A Admitting a new domain

SLOWLAB is a harness for a class of problems rather than for one domain. A domain is admissible when it supplies four things.

The conditions must hold jointly and by nature, not because the benchmark author chose hyperparameters that way (§4.2).

A published forward model must exist, so that the response surface, its noise structure and the location of its optimum are fixed outside the benchmark author’s control.

A cost model in the objective’s own units must be available, so that exploration is priced rather than constrained from outside.

Experimental units must not be interchangeable. Some structure—blocks, batches, chambers, cohorts—must constrain which treatments can be assigned where, because that is what makes allocation a decision rather than bookkeeping, and what makes replication compete with coverage.

Greenhouse horticulture, the instantiation built here, satisfies all four (§4.2). Microbial evolution experiments, bioprocess scale-up, materials ageing and dose-finding trials meet them with published forward models already in hand. We do not claim results transfer across domains—that is the question a multi-domain version would exist to answer—only that the harness, the measures and the reference points are defined at the level of the class.

Four extensions that need no environment code. The crop model already contains the mechanism each of them tests, so each is a task specification. *Equipment faults* give ground-truth attribution: a heater fails in one chamber for k days, the agent sees an unexpectedly low yield and an anomalous log, and must name the cause, which is exact because we injected it. *Fruit abortion* above T_{CRIT} already makes irreversibility graded rather than binary, from recoverable through partially recoverable to a terminated unit. *Between-season parameter drift* within the published between-site spread makes conclusions expire on a schedule the data rather than we set. And because the oracle is computable in a tenth of a second, a *target profit rate* sorts instances into comfortably feasible, feasible only in a narrow region, marginally infeasible and clearly infeasible, which tests whether an agent can recognise that it has been asked for something impossible. None of these can be built on a surrogate without injecting the abstraction by hand.

B A worked round

Figure 3 runs one round of OPTIMISE through in full: one site, one twelve-unit budget, two designs that both pass the feasibility check and differ only in how the units are allocated. It is the smallest case in which the control hierarchy of §3.2.3 decides the answer, and the reason the environment scores allocation rather than treatment choice alone.

C Task specifications

Factor	Range	Control level	Role
day temperature	18–32 °C	chamber	heating/ventilation setpoint
night temperature	12–22 °C	chamber	setpoint
CO ₂	350–1200 ppm	chamber	enrichment
PAR	18–32 mol m ^{−2} d ^{−1}	chamber	natural + supplemental
plant density	2.2–4.5 m ^{−2}	loop	planting decision
LAI _{max}	2.0–5.0	loop	pruning target

Table 6: The six management factors. Control level is a physical fact—lamps and heating serve a whole chamber, whereas density and the pruning target are set per hydroponic loop—and it determines each factor’s parallelism ceiling.

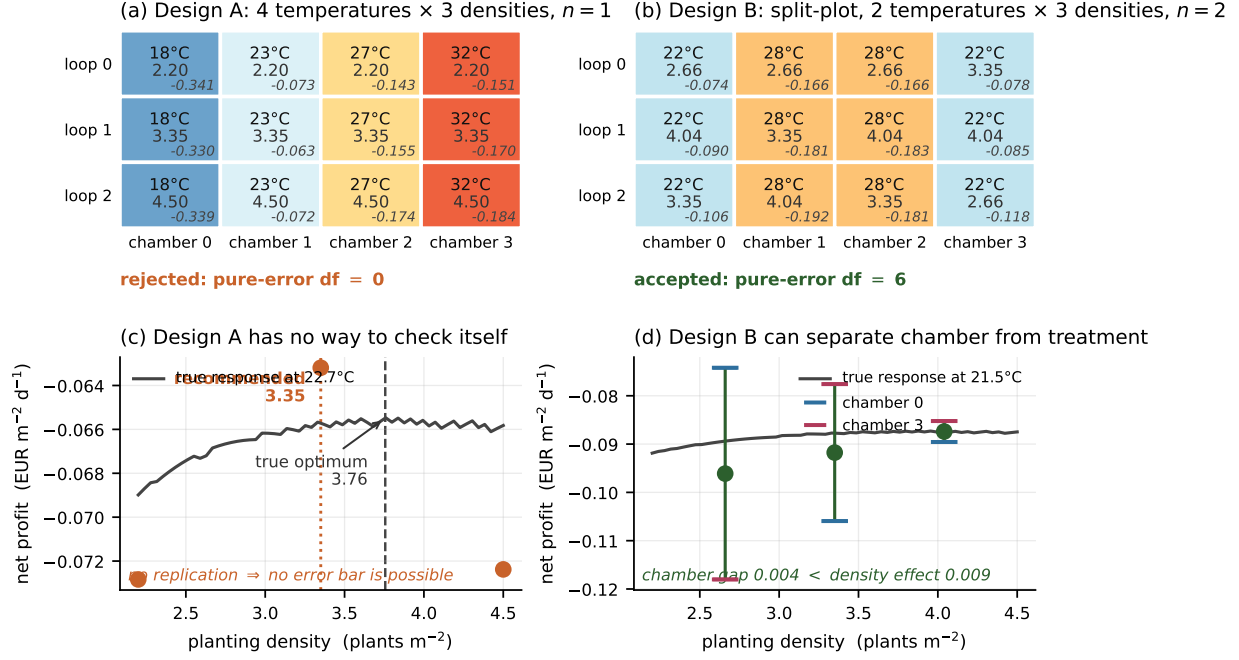


Figure 3: One round of the running scenario: same site ($\text{seed}=0$), same twelve-unit budget, two designs. All values are produced by the environment; the script is in the repository. (a) The intuitive design fills the grid—four temperatures, one per chamber, crossed with three densities. It passes `validate_design`, but every treatment is observed once, so pure-error df is zero and each temperature is confounded with one chamber. (b) The split-plot: two temperatures, each replicated across two non-adjacent chambers, densities randomised within chamber. Our first attempt used chambers 0–1 and 2–3 and was rejected by the randomisation check—temperature then correlates perfectly with chamber index. (c) Design A’s best observation sits at $3.35 \text{ plants m}^{-2}$ where the true optimum at that temperature is 3.76 ; the miss is small, and the point is that with one observation per treatment nothing in the data says whether it is small. (d) Design B can answer the question it was given: the chamber 0 against chamber 3 gap at identical settings is 0.004 against a density effect of 0.009 , so the effect is resolvable above the block noise — which is a statement design A cannot make either way. The figure is regenerated from the frozen environment; an earlier version hard-coded the annotations and printed the comparison in (d) with the inequality facing the other way.

The noise column of Table 2 is

$$\sigma_{\text{tot}}/\sqrt{B} = \sqrt{\text{sd}^2\left(\frac{\tau_{\text{loop}}^2}{B} + \frac{\tau_{\text{chamber}}^2}{CR} + \frac{\tau_{\text{batch}}^2}{R}\right)},$$

the standard error of the mean of $B = R \times |\mathcal{U}_r|$ observations spread over C chambers and R rounds. Only the first term falls with B : a chamber effect is shared by every unit in that chamber and a batch effect by every unit in that round, so past a point buying more units stops buying precision. That is the width constraint of §3.1 in arithmetic.

Two quantities a grower does control are declared by the task rather than exposed as factors, and the environment refuses to treat either as one. Cycle length is fixed at 210 days: it is monotone over any plausible range, because fruit set does not begin until roughly 30–45 days and that establishment period is pure cost, so exposing it would add a dimension with the same answer at every site. Leaf removal is fixed at 2.25 g node^{-1} because the reduced model gives defoliation no photosynthetic penalty, so its optimum is always at the upper rail. Both are asserted rather than silently overridden, so anyone who adds them back gets an error instead of a sweep that quietly holds them constant.

D The naive corner

Setting every factor to its upper bound yields *negative* profit at every site, worse than the site optimum by a factor of 4 to 52; setting every factor to its midpoint is also negative at most sites, worse by a factor of 1.3 to 21. Maximising everything fails because day temperature at its ceiling aborts fruit set (eq. 8) while supplemental lighting continues to draw power. We use this rather than “every factor has an interior optimum” as the criterion for a non-trivial search problem.

E Reproducibility

All results use seeds $0..N-1$; `SlowLabEnv(task, seed)` is deterministic, and the seed selects a sampled site rather than only a noise realisation. Every table and figure in this paper is emitted by a script in `scripts/` from the JSON under `results/`, and none is hand-edited. The files carry the environment version in their names, so a table built from a previous environment cannot be picked up silently. Code, task definitions and those scripts are released under the MIT licence at <https://github.com/dingningpei/slowlab-benchmark>. The 800 raw language-model transcripts and the cached atom sets are published there as a release asset rather than in the repository itself. **[TODO: datasheet, croissant metadata, hosting]**

F The $\mathcal{R}^*$ estimator

Observation likelihood. Because the per-unit draw ϕ_u is pushed through the forward model, $p(y \mid \theta, x)$ has no pointwise density. We moment-match the per-unit response to a Gaussian with variance $s^2(x)$, which makes the covariance of a round’s $|\mathcal{U}_r|$ observations

$$\Sigma(D_r) = \text{diag}(s^2(x_u) + \tau_\gamma^2) + \tau_\beta^2 Z Z^\top + \tau_\epsilon^2 \mathbf{1}\mathbf{1}^\top,$$

with $\tau_\beta, \tau_\gamma, \tau_\epsilon$ the nuisance scales of §3.1 and $Z \in \{0, 1\}^{|\mathcal{U}_r| \times C}$ the block-membership matrix. The allocation enters through Z , so where a design places its treatments changes (6), not only which treatments it chooses.

Three approximations. (a) The Gaussian moment-match above. (b) $s^2(\cdot)$ is computed once at the nominal parameter vector rather than per atom. (c) The prior is the M -atom mixture rather than the continuous P_Θ . We had argued that (c) was benign, on the grounds that the dimension which matters is that of x_θ^* rather than of θ , and that it is small — over 150 instances of OPTIMISE the true optima occupy ten grid cells at $H[p(x_\theta^*)] = 2.05$ nats. That argument is wrong, and the next two paragraphs are what it cost. The support of x_θ^* being small does not make the *likelihood* over atoms well behaved, and it is the likelihood that fails. **[PENDING: Moment diagnostics for (a) and (b).]**

The in-sample estimator and why we abandoned it. Our first estimator drew one atom set and used it for everything: the posterior weights $w_m(y)$, the minimisation over candidate recommendations, and the evaluation of the chosen recommendation. The failure is mechanical. $\Sigma(D_r)$ above contains observation noise and block structure and nothing for the coarseness of the grid, so with 12 to 24 observations the likelihood discriminates between atoms far more sharply than the grid spacing warrants. Measured over 160 atoms on OPTIMISE, the mean effective sample size $1/\sum_m w_m^2$ is 1.1, and the candidate that minimises $w^\top D$ is some atom’s own optimum in 98% of draws. The estimator was therefore answering “can the data identify which of these 160 sites generated it” and reporting the answer as though it were “how much decision risk does this design remove”. Held out on atoms that did not generate the data, the same designs score $c = -64\%$ where the in-sample number reads $+97\%$: conditioning on the design leaves the reasoner worse than the prior-optimal constant, because the posterior concentrates on a wrong atom and the recommendation follows it there.

The estimator we report. Three changes, all in `achievable_regret_cv`. The likelihood is raised to the power $1/T$, equivalently $\Sigma \rightarrow T\Sigma$, which is the minimal way to state that the true site lies between atoms. T is selected by leave-one-out over the inference atoms: for each atom m , draw y from m , form the posterior from the others, choose a recommendation, and score it at m ; take the T minimising the average.

The evaluation atoms are never touched, so selecting T cannot import the bias it removes, and the leave-one-out curve tracks the held-out curve closely enough to be used as its proxy. Finally the recommendation is scored on the disjoint half. The result is c between 1% and 20% where the in-sample estimator read 35% to 78%, and the orderings differ: on OPTIMISE the in-sample estimator ranked random search first among the reference strategies and the repaired one ranks it last.

We tried the obvious cheaper fix first. Fitting the surrogate hyperparameters is not it and neither is raising M alone; tempering is what moves the number, and the temperature the calibration selects is 3 to 100, meaning the effective observation covariance is one to two orders of magnitude larger than the noise model alone implies. That is the size of the discretisation error, stated in the only units available.

Convergence in the atom count. Table 7 runs seven designs at $M = 80$ and $M = 320$ on the two tasks we could afford at both counts, six runs each. The designs span the range of c rather than compete: space-filling, a two-level factorial, four, two and one distinct treatments, and the two degenerate cases of every unit at a corner and every unit at the centre.

The estimator has not converged and the direction of its error is now the opposite of the in-sample one. Quadrupling M leaves the degenerate designs where they are and lifts the good ones: on SCREEN the space-filling design goes from 0.8% to 16.5%. The reason is the split. At $M = 80$ only 40 atoms carry the posterior, which is too coarse a stand-in for P_Θ for any design to separate, so the measure reports that no design helps. Every level we report is therefore a lower bound on what a converged estimator would give, which is the conservative direction for claims of the form “no agent reaches the ceiling” and the awkward one for any claim that a particular agent’s design was poor.

Rank correlation between the two counts is 0.82 on OPTIMISE and 0.64 on SCREEN. That is lower than the in-sample estimator reported and it is the honest number: the in-sample estimator’s near-perfect rank stability was itself an artefact, since a posterior that always collapses onto one atom is a deterministic function of design geometry and therefore stable in M for the wrong reason. A single estimate carries a standard deviation of 2 to 4 points. Every c in this paper is a mean over dozens of designs, where that falls below one point, and the comparisons we draw conclusions from are paired at fixed M .

Design	OPTIMISE		SCREEN	
	c at $M=80$	$M=320$	c at $M=80$	$M=320$
space-filling	21.7 ± 1.3	29.6 ± 2.0	0.8 ± 0.9	16.5 ± 0.8
four treatments	2.2 ± 1.6	16.0 ± 1.4	0.5 ± 1.1	7.8 ± 0.8
two treatments	4.4 ± 1.5	12.2 ± 1.5	12.9 ± 0.9	7.8 ± 1.1
two-level factorial	-2.0 ± 1.8	-0.5 ± 2.1	-1.7 ± 0.4	3.4 ± 1.1
corner only	-1.9 ± 1.5	-1.7 ± 2.1	0.7 ± 0.8	1.5 ± 0.8
centre only	-0.4 ± 1.6	-1.9 ± 1.7	-0.4 ± 1.3	-3.1 ± 0.5
one point	-2.0 ± 1.6	-2.3 ± 2.1	-0.7 ± 1.1	0.6 ± 0.6
$\mathcal{R}^*(\emptyset)$: OPTIMISE $0.00546 \rightarrow 0.00610$; SCREEN $0.00500 \rightarrow 0.00443$				
rank correlation of c between the two atom counts: 0.82 and 0.64				

Table 7: Design efficiency at two atom counts under the cross-validated estimator, seven designs chosen to span the range rather than to compete. Each entry is the mean of six independent runs with its standard error; a single run carries a standard deviation of 2 to 4 points, which is why the levels reported elsewhere in this paper are always means over dozens of designs. Quadrupling the atom count leaves the two degenerate designs where they were and roughly doubles the best one on OPTIMISE; on SCREEN it moves the space-filling design from nothing to the top. The estimator is not converged, and the direction of its error at low M is to under-credit good designs, because half of 80 atoms is too coarse a stand-in for P_Θ for any design to separate.

The bound does not survive estimation. For the exact quantities $c \in [0, 1]$ is a theorem: numerator and denominator start from the same beliefs, so $\mathbb{E}[\min] \leq \min[\mathbb{E}]$ gives $\mathcal{R}^*(D_r) \leq \mathcal{R}^*(\emptyset)$, and $\mathcal{R}^* \geq 0$ closes the other end. The estimator above is a tempered rule graded out of sample, about which the optimality

theorem says nothing, and the difference is visible: across the 320 reference rounds behind Table 8, 69 carry a negative c . Averaged over the dozens of rounds that make up one cell every value we report is inside $[0, 1]$ — c from 1% to 20%, η from 4% to 56% — but by margin rather than by proof. For η there is not even an exact bound to appeal to, since the agent’s $\hat{x}_r$ has seen every round while the numerator scores each round separately against the prior.

Trajectory and endpoint. The interim recommendation elicited after each round uses the same rule the agent applies at the end, so the last point of a trajectory equals the regret of the submitted recommendation. This is asserted by a regression test: an earlier version used the best observed point for the trajectory and a posterior maximum for the submission, which differed by 47% on OPTIMISE.

G The horticultural instantiation

Forward model. f_θ is the reduced state-variable TOMGRO model [Jones et al., 1999]: five coupled state variables—mainstem nodes, leaf area index, above-ground, fruit and mature-fruit dry weight—implemented from its equations (1)–(11) with the published least-squares estimates. We deliberately do not sample ground truth from a Gaussian-process prior: such a sample has no physical content, its length scale would be our choice, and it makes the modelling assumption of GP-based agents exactly correct. Fitting our Bayesian-optimisation reference’s own surrogate to the mechanistic response gives mean absolute error $1.1\times$ the global average in the high-temperature decline and $2.2\times$ near the kink where supplemental lighting begins to incur cost—real but mild, and the surrogate still locates the optimum. What the mechanistic surface supplies that a stationary GP cannot is structure: fruit abortion above T_{CRIT} (eq. 8), the source of irreversibility, and a source-sink limit taking the *minimum* of two growth expressions (eqs. 9–10). P_Θ follows the same paper’s finding that vegetative parameters are stable across sites while fruit parameters vary: each seed draws fruit parameters across the published between-site spread (N_{FF} varies by $2.2\times$ across their three sites). Five of the six factors have optima that move across sites; the sixth, supplemental light, sits at the lower rail in every instance because at 2024 European electricity prices it never repays its energy.

Facility. Blocks are climate chambers and units are hydroponic loops. Lamps, heating and CO_2 enrichment are set per chamber; planting density, pruning target and the pruning target is set per loop. Cycle length is fixed at 210 days for every unit and is not a decision (§4.3).

Costs. Revenue is fruit fresh weight at wholesale price; costs are supplemental lighting, heating and CO_2 enrichment, computed with standard greenhouse engineering relations. Coefficients come from public sources rather than from us, because the *Costly* measure is denominated in euros and an uncited coefficient makes its absolute value uninterpretable: crop price 1.50 EUR/kg fresh weight, between Dutch (1.57) and Spanish (1.35) vine tomato in March 2024 [HortiDaily, 2024]; electricity 0.189 and natural gas 0.0605 EUR/kWh, the EU non-household averages for 2024 [Eurostat, 2025]; luminaire efficacy $2.5\mu\text{mol J}^{-1}$, the level over half of listed horticultural fixtures reach [DesignLights Consortium, 2022]; a cover heat-transfer coefficient of $4.0\text{ W m}^{-2}\text{K}^{-1}$, between single glass (≈ 4.5) and double polyethylene (2.9–3.4) [Kittas and Baille, 1998]. The CO_2 leakage coefficient yields 14.6 kg m^{-2} per season, inside the 12–24 range that must be delivered for tomato [AHDB, 2020].

Caveats. The figures are representative European values, not measurements from one facility. Outdoor temperatures correspond to the Mediterranean-subtropical winters of TOMGRO’s calibration sites rather than to northern Europe, where greenhouses consume $47\text{--}69\text{ m}^3$ of gas per m^2 per year ($\approx 460\text{--}670\text{ kWh}$) against about 245 here—a difference of climate rather than of coefficients. And one consequence was not our choice: at these prices supplemental lighting is never economical in this climate, so the optimum for PAR sits at its lower bound for every site.

H Scripted references, and how far they get

Four scripted strategies ship with the environment: random search over the factor space; a planned campaign, being a Plackett–Burman screen with centre points followed by a response surface, replicated and randomised

Task	Strategy	$\bar{\mathcal{R}}(R) \downarrow$ final answer	$\mathcal{R}_{\text{cum}} \downarrow$ campaign cost	$c \uparrow$	$\eta \uparrow$	design rank	decision rank
SANITY	$\mathcal{R}^*(\emptyset) = 0.0055$						
	Random search	0.0417(119)	345	1%	13%	3	4
	Planned campaign	0.0089(33)	140	9%	56%	1	1
	Bayesian opt. (1 rep)	0.0172(73)	289	1%	32%	3	3
	Bayesian opt. (2 rep)	0.0128(49)	288	5%	41%	2	2
SCREEN	$\mathcal{R}^*(\emptyset) = 0.0044$						
	Random search	0.0585(92)	1061	20%	6%	1	1
	Planned campaign	0.0989(84)	972	5%	4%	4	3
	Bayesian opt. (1 rep)	0.0586(183)	799	16%	6%	2	1
	Bayesian opt. (2 rep)	0.0785(188)	844	12%	5%	3	2
OPTIMISE	$\mathcal{R}^*(\emptyset) = 0.0061$						
	Random search	0.0258(82)	709	3%	23%	3	4
	Planned campaign	0.0251(98)	510	3%	24%	3	3
	Bayesian opt. (1 rep)	0.0120(37)	496	6%	48%	1	2
	Bayesian opt. (2 rep)	0.0107(37)	540	5%	54%	2	1
TRANSFER	$\mathcal{R}^*(\emptyset) = 0.0041$						
	Random search	0.0706(51)	2404	17%	5%	1	4
	Planned campaign	0.0653(81)	1592	8%	6%	4	3
	Bayesian opt. (1 rep)	0.0335(36)	1418	13%	11%	3	1
	Bayesian opt. (2 rep)	0.0534(152)	1644	15%	7%	2	2

Table 8: Reference strategies, environment v1.0.0, eight instances per task. $\bar{\mathcal{R}}(R)$ is the regret of the final recommendation, with its standard error in the last two places; $\mathcal{R}_{\text{cum}}$ is the profit the campaign forwent, in € per square metre summed over committed unit-days. c is design efficiency against the $\mathcal{R}^*(\emptyset)$ printed above each block. Rank 1 is best and tied values share a rank; decision rank orders η , whose levels are in Appendix F. We draw conclusions from the ranks rather than the levels, for the reason given in §3.3.2.

across chambers; and Bayesian optimisation at one and at two replicates per treatment. Table 8 runs all four on all four tasks, eight instances each. They are a reference point, not a baseline we claim to have tuned: each is a textbook method implemented straightforwardly, and the numbers below say what such an implementation achieves at these budgets, not what the method is capable of.

What is and is not tuned. The Bayesian-optimisation reference uses a hand-rolled Gaussian-process surrogate whose length scale and observation noise are fixed rather than fitted, and the tool arm of §4.4 hands the same routine to a language model. We assumed the fixed hyper-parameters were costing it and checked (`scripts/tool_hyper_control.py`): refitting length scale and observation noise by marginal likelihood on exactly the same observations, holding the recommendation rule and everything else constant, changes the regret of the point proposed by -25% on OPTIMISE, -6% on SCREEN, -5% on TRANSFER and $+32\%$ on SANITY: better on three tasks, worse on one, and still several times the regret the same routine carries when it chooses its own designs. The fixed hyper-parameters are not what limits either. The planned campaign’s screen is full rank over the factors it varies, which is the only property we checked. Both are released so that a reader can improve on them.

The two efficiencies come apart on TRANSFER. Random search ranks first on design there and last on decision: it spreads its units over the whole space, so its observations locate the optimum better than any other strategy’s, and it then recommends close to the best point it happened to run. Bayesian optimisation is the mirror image, concentrating units near the incumbent and converting far more of what that carries. On SCREEN the same pattern holds weakly. It does *not* hold on OPTIMISE, where the repaired estimator puts all four strategies between 3% and 6% and the task no longer separates designs at all; the in-sample estimator reported 39% to 46% there and ranked random search first, and that was an artefact. On SANITY the planned campaign leads on both, which is what a control should look like. The stronger demonstration

that the two measures are not restatements of each other is in §4.4, where one intervention moves them in opposite directions within a single model.

Where the ratio is thin, nothing converts. On SCREEN all four strategies convert between 4% and 6%, so the decision column there is close to two tied pairs rather than an ordering. This is Table 2’s ratio of 1.5 restated: when the room for tailoring is barely above what a full budget resolves, the difference between a good and a bad reading of the data is not detectable.

A screen presumes a campaign these budgets do not pay for. On SCREEN the planned campaign has the worst realised regret of the four, worse than random search, although its screening design is full rank over the six factors it varies, replicated and randomised across chambers. At this budget it spends half its units establishing which factors matter and the other half on a response surface around a noisy incumbent, while random search spends everything on coverage. Plackett–Burman was designed for a programme with a follow-up; two rounds are not one.

I How much of the ceiling any design can remove

§4.4 reports design efficiency without saying what value of it is attainable, and a reader is entitled to ask whether a small c measures the agent or the task. If no admissible design can push $\mathcal{R}^*(D)$ far below $\mathcal{R}^*(\emptyset)$, the denominator of (7) carries no information and a low score says only that the problem is hard.

We answer by searching the admissible set directly (`scripts/oracle_design.py`). The search reads P_Θ and the forward model, so it is not an agent and its designs are not achievable by one: it is an upper bound on what the design half of the score could ever report. The set is discrete and hierarchical — chamber-level factors take one value per chamber — so a round is a choice of how many chambers, how many loops in each, how many distinct chamber-level and loop-level settings, and where those settings sit in the factor space. We enumerate that grid over five standard point families (maximin, Latin hypercube, two-level corners, three-level grid, uniform), check every candidate through the environment’s own validator, and score it with the cross-validated estimator of §3.3.2. Being an enumeration over families rather than a proof of optimality, the $c_{\max}$ of Table 9 is itself a lower bound on the true maximum.

Task	$\mathcal{R}^*(\emptyset)$	best design found		best c achieved		median c at	
		$c_{\max}$	k	model	scripted	$k = 2$	$k \geq d+1$
SANITY	0.0055	31.7%	3	11.5%	9.2%	1.4%	5.3%
SCREEN	0.0044	27.8%	5	7.1%	20.0%	2.0%	10.7%
OPTIMISE	0.0061	37.4%	9	18.0%	5.5%	9.9%	15.6%
TRANSFER	0.0041	26.5%	8	9.2%	16.8%	4.6%	12.5%

Table 9: What one round of design can buy. $c_{\max}$ is the highest design efficiency found over the admissible set, and k the number of distinct treatments the design carrying it uses. The next two columns are the best c any language model and any scripted strategy reached on that task. The last two are the *median* c over all candidates with two distinct treatments and over all candidates with at least $d+1$, which is the comparison the diagnosis in §4.4.2 rests on.

Three readings. First, **the ceiling is not approachable**: the best design found removes 27% to 37% of $\mathcal{R}^*(\emptyset)$, so most of the no-data risk survives any single round at these budgets. That is a property of the regime rather than of the agents — twelve to twenty-four units against a six-dimensional site distribution cannot identify the optimum — and it is a second reason, beyond the estimator’s convergence, to read c as a rank rather than as a fraction captured.

Second, **the headroom over what agents reach is real**. The best c anyone achieves is $1.4\times$ to $2.8\times$ below $c_{\max}$ on every task, so the low values in Table 3 are not the measure bottoming out.

Third, **the width diagnosis survives its own counterexample**. The best design we find on SCREEN uses five distinct treatments over six factors, so it is rank-deficient by the criterion of §4.4.2: a well-placed narrow design can do well. But the median tells the other story — at $k = 2$ the median candidate captures

1.4% to 9.9% depending on the task, against 5.3% to 15.6% at $k \geq d + 1$, and the ordering holds on all four. An agent choosing a narrow design is choosing from that distribution without being able to see the surface in advance, and the agents’ realised c sits close to the narrow median rather than to its best case.

J Does the ordering replicate?

Two tasks have a prize no larger than the noise a full budget leaves behind (Table 2: the ratio is 0.8 on SCREEN and 1.0 on TRANSFER), which invites the objection that any ordering they produce is a coin toss. We measure it (`scripts/rank_stability.py`). The twenty instances are split into two disjoint halves of ten, the five models are ranked on each half by mean realised regret, and the two rankings are correlated; every model sees the same sites within a half, so the only thing varying is which sites were drawn. Over 4000 random splits:

Task	ratio (Table 2)	ρ between halves	$P(\rho > 0)$	same winner
SCREEN	0.8	−0.28	0.27	0.09
TRANSFER	1.0	+0.09	0.54	0.23
SANITY	1.3	+0.40	0.83	0.42
OPTIMISE	2.2	+0.76	1.00	0.12

Table 10: Split-half stability of the five-model ordering by realised regret, bare arms, ordered by the ratio of Table 2. *same winner* is how often the best-ranked model agrees between halves.

The ratio predicts the answer exactly: in Table 10 the four tasks fall in the same order on both columns. At these sample sizes only OPTIMISE produces a per-model ordering that replicates; SCREEN’s anti-correlates and TRANSFER’s is indistinguishable from zero. The design and decision efficiencies behave the same way (c at −0.13, +0.55, +0.36 and +0.74 in the same task order), which is expected, since they are computed on the same twenty draws.

This bounds what §4.4 may claim. Every finding there that names an individual model on SCREEN or TRANSFER is under-powered, and we do not make one. What the section does claim is of three other kinds, none of which ranks models: proportions of submitted designs, which do not involve the outcome at all; within-model contrasts paired by site and tested as such, which is how the tool arm is read; and statements that hold in every cell rather than in a particular order among them. Raising the instance count is the obvious remedy, and the cost is linear in it: twenty sites times four tasks times ten arms is already 800 episodes.

K Pricing the irreversibility axis

Three of the four conditions have no dial in this domain: a growing cycle takes as long as it takes, plant-to-plant variation is what it is, and a crop grown is a crop sold. The fourth has one. Commitment is irreversible throughout — a submitted design cannot be revised and the units it occupies are gone for a cycle — but beyond commitment the environment implements damage: fruit abortion above T_{CRIT} puts a unit into a recovery period graded by how far the threshold was exceeded, so a badly chosen treatment costs capacity in the *following* round and not only in its own. The four tasks ship with that recovery length set to zero, so every number in the main tables is at zero. This appendix turns it up, which is what makes “the four conditions matter jointly” a measurement rather than a claim about the domain.

It is not dead code for want of triggering: across the reference strategies 55% to 60% of committed units exceed the threshold, by 4.3 to 4.6 °C on average.

Scripted references. Each of the four is run on each task at recovery lengths of zero, sixty and two hundred and ten days, twenty instances each, identical in every other respect and paired by site (`scripts/recovery_sweep.py`). The dial bites: at sixty days every episode but one of the 320 loses capacity, and the cell means run from 78 to 1071 unit-days, 2% to 12% of what the campaign used; at two hundred and ten they run from 272 to 3747, 8% to 42%. Realised regret moves by −10% to +20% and *not one of the sixteen cells moves by as much as two standard errors of its own paired difference*. The two non-zero settings

are also indistinguishable from each other, in every cell to four digits, because the blocking granularity is a round: both remove the unit from exactly one.

Language models. That measurement has a boundary: none of the four references replans around lost capacity, since each constructs its design from the task rather than from what the facility can still run, so it prices the loss of units to an agent that ignores it. A language model does see the shortened `available_units` list and can respond, so we ran that arm too (`scripts/recovery_llm.py`): five models on OPTIMISE, the same twenty sites, bare interface, 210 days against the 0 of the main tables, one hundred paired episodes. The dial reaches the agents — the rounds they submit shrink by 27% and 23% in the second and third rounds, and infeasible submissions per episode rise from 0.70 to 1.22 as models keep asking for units the facility no longer has. Realised regret again does not move: 0.0230 against 0.0221, a paired t of -0.34 at $p = 0.74$, worse in fifty-one of the hundred. Per-model differences run from -33% to $+44\%$ with one at $p = 0.02$, but at a sampling temperature of 0.7 a rerun at the same setting would move a twenty-instance cell by about as much (Appendix J), so only the pooled test is worth reading.

Both halves say the same thing, and it refines the claim rather than supporting it. At these budgets extra units buy nothing, so losing them costs nothing; what binds is the number of cycles and the width of the design, not the number of slots. Irreversibility in v1.0 therefore acts through *commitment* — a design cannot be revised and a cycle is spent either way — and not through the loss of capacity, even when a quarter of the facility is withdrawn. A version that wanted capacity loss to bite would have to make replication pay first.

L Known defects, and why we stopped fixing them

The environment is frozen at v1.0. This register is complete as far as we know it, and we do not intend to fix all of it. A benchmark is a fixed artefact; its value is comparability across agents, not correctness of the simulator.

The stopping rule. *A defect is fixed if it changes a claim the paper makes, and recorded if it does not.* We adopted this after the alternative became visible: while $\mathcal{R}^*(\emptyset)$ was being used as an admission test for tasks, every correction to the crop or cost model moved the ceiling, which moved the task set, which motivated the next correction. That loop does not terminate, because a simulator can always be made more nearly right. Reporting ranks rather than levels (§3.3.2) removes the dependence and lets the rule bite.

The rule caught four defects and let three stand. Four defects changed claims and were fixed: a weight λ on the cost term, nominally a task-design parameter, which set where every optimum sat and pushed four of six factors onto their rails at $\lambda = 0.25$; a photosynthesis routine that read global nominal parameters instead of the site’s, so that every site had an identical light and CO_2 response and a stated limitation about sampling those parameters was a no-op; a heating model that applied $AU \Delta T$, the peak-sizing formula for the coldest night, to all 24 hours of 210 days, overstating a term worth 48% of variable cost; and an objective named net profit that omitted harvest labour, grading loss and all fixed costs. One defect was recorded instead: the block effects below, which affect only the ratio in Table 2, and that ratio is no longer a gate.

Noise model. The block scales τ_{chamber} , τ_{loop} and τ_{batch} multiply the standard deviation of the objective over uniformly random points in the whole factor space, a quantity dominated by corners no grower would operate in. Improving the ventilation physics left the optimum unmoved to three figures and inflated that standard deviation by 71%, and with it the chamber effect, from 4.4% to 7.9% of revenue at the optimum. A chamber’s climate uniformity has nothing to do with how bad the worst corner of the design space is; the effect should be a temperature offset pushed through the forward model, with a magnitude taken from measured greenhouse uniformity. Separately, `task.noise_sd` is a declared value 1.3 to 1.9 times the per-unit standard deviation the simulator actually produces; the simulator is right and the declared value is unused except in reporting.

Crop and cost model. The profit-optimal yield averages $28 \text{ kg m}^{-2} \text{ yr}^{-1}$ over sites, range 21 to 37, against a commercial 50–70, while the model’s *maximum attainable* yield is 59, inside that range: the model can grow a commercial crop and does not think it pays. We have not established why. Supplemental light is rail-bound at every site, so the entire lighting sub-model has zero elasticity on the objective. This is also what lowers the SCREEN ceiling, and it separates two questions a sensitivity analysis conflates: over 24 sites, setting supplemental light mid-range rather than at its optimum costs 0.080, the largest of the six factors, while setting it at the across-site mean optimum — all an experiment could still recover — costs 0.000, because its optimum does not move. The factors worth an experiment are day temperature and CO_2 , at 0.0085 and 0.0060, together 85% of what instance-tailoring can recover there (`scripts/factor_value.py`). Relatedly, $\text{par}_{\text{ambient}}$ is fixed across sites although natural daily light integral is the largest real difference between a Dutch and an Almerian greenhouse. A daily-timestep model cannot represent a time-varying CO_2 enrichment strategy, which is what growers actually run. The ventilation capacity is clipped rather than allowed to fail, so overheating above that cap is not modelled.

Provenance of coefficients. Electricity and gas prices, the site-level spreads, the LED efficacy range and the cover heat-transfer range are cited. Four numbers are not: the tomato price, whose source we can no longer verify and which carries the largest elasticity on the objective; the fruit dry-matter fraction, which has no citation at all and carries the second largest; the transplant cost, quoted in dollars and used as euros; and two labour coefficients whose derivations contain wage and task-share figures that are ours. One property limits the damage: the tomato price and the dry-matter fraction scale revenue linearly and $\mathcal{R}^*$ scales with them, so they cancel from every ratio we report. $\mathcal{R}^*(\emptyset)$ is 3.1% of revenue at the optimum on both SANITY and OPTIMISE, and adding harvest labour and grading loss left that ratio where it was. These coefficients set the euro scale and no conclusion.

Estimator and action space. The estimator’s own defects, and the one we found and repaired, are in Appendix F; they are the largest single item in this register. Separately, replication does not pay at these budgets, so an agent is never rewarded for buying precision — an arithmetic consequence of bounded parallelism rather than a bug, but it narrows what the benchmark can distinguish.

A stronger form of irreversibility is implemented and not enabled. All four tasks set the recovery length to zero, with the comment “0 = off (default, keeps existing results unchanged)” — which is the reasoning this appendix opens by arguing against, and we record it as such rather than defend it. Appendix K turns it on and reports what changes.

Two actions a real grower has and this agent does not. The action space stops at a design and a recommendation, and two of the omissions matter more than the rest.

Early termination. A greenhouse is watched continuously—climate at minute resolution, crop registration weekly, harvest accumulating from about week ten—so a grower who has planted a bad treatment usually knows by week six and pulls it. We have no such action: a committed unit runs its full 210 days. The omission is what makes §3’s two readings of delay, information and action, coincide here; restoring it would separate them, and it is the cleanest available trade-off between slowness and irreversibility, since salvaging costs part of a season rather than all of it.

Staggered planting. Rounds are synchronous: every unit in a round starts and ends together. Real trial programmes run a pipeline, starting a new batch every few weeks. Our budgets leave half the facility idle on two of the four tasks—8 of 16 unit-slots on SANITY and 12 of 24 on SCREEN—for the full 420 days, which no grower would accept. Staggering at a 105-day offset would run four batches instead of two in the same wall-clock. It also poses the harder question: with only partial information from batch one, what do you plant in batch two? That is where slow feedback actually bites, and it is the first thing we would add.

An agent also cannot choose what to measure, when to measure it, or how many plants to grow per unit.